Monday warm-up: unit analysis and vectors

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1 Chapter 1 - Unit analysis, Estimation

1. Estimate the area of the North Quad of Whittier College (the open space outside the SLC):
   - A: 5000 m^2
   - B: 5000 cm^2
   - C: 500 m^2
   - D: 500 cm^2
2. A coffee bean is about 0.5 cm^3 in volume. How many could fit in a 2 liter bottle?
   - A: 4×10^1
   - B: 4×10^2
   - C: 4×10^3
   - D: 4×10^4
3. What is 25 m s^{-1} in km hr^{-1}?
   - A: 15 km hr^{-1}
   - B: 25 km hr^{-1}
   - C: 60 km hr^{-1}
   - D: 90 km hr^{-1}
4. Suppose a train leaves Los Angeles heading North to San Jose. (a) If the distance is 375 miles, what is this in kilometers? (b) If the journey takes 8 hours, what is the average speed of the train? (c) If the journey was instead 4 hours long, what would be the speed?
5. Suppose a ship accelerates from 0 km hr^{-1} to 10 km hr^{-1} in 60 seconds. If acceleration is the change in velocity divided by the change in time, what is the acceleration of the ship?

2 Chapter 2 - Vectors

1. A **vector** is a physical quantity with a magnitude and a direction: $\vec{v} = v_x\hat{i} + v_y\hat{j}$. The v_x is the amount in the x-direction, and the v_y is the amount in the y-direction. The $\hat{i}$ and the $\hat{j}$ are **unit vectors**. The $\hat{i}$ points in the x-direction, and the $\hat{j}$ points in the y-direction. Each has length 1 and no units. (a) Let $\vec{v} = -2\hat{i} + 2\hat{j}$, and $\vec{w} = 2\hat{i} - 2\hat{j}$. Draw each in a 2D coordinate system below. (b) What is $\vec{v}+\vec{w}$? (c) What is $\vec{v} - \vec{w}$?